

left with 31 white and 31 black squares. We shall tile the checkerboard using 31 dominoes in the following manner:

Domino 1: Row 2 Column 1, Row 3 Column 1

Domino 2: Row 4 Column 1, Row 5 Column 1

Domino 3: Row 6 Column 1, Row 7 Column 1

Column 2: Row 1, 2 → Domino 4 Row 3, 4 → Domino 5 Row 5, 6 → Domino 6 Row 7, 8 → Domino 7

Column 3: ~~Row 1, 2 → Domino 8~~ In a similar manner as Column 2, we use Dominoes 8 to 11

Column 4: Dominoes 12 to 15 (Similar to Column 2)

Column 5: Dominoes 16 to 19 (Similar to Column 2)

Column 6: Dominoes 20 to 23 (Similar to Column 2)

Column 7: Dominoes 24 to 27 (Similar to Column 2)

Column 8: Dominoes 28 to 31 (Similar to Column 2) □

44. Prove or disprove that you can use dominoes to tile a standard checkerboard with all four corners removed.

Solution: A standard 8x8 checkerboard has 32 black squares and 32 white squares. On removing 4 corners we are left with 30 black squares and 30 white squares. We shall tile the checkerboard using 30 dominoes in the following manner.

B	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W